

L12: Lesson Summary



This lesson focused on network models.

Some models are simple, with only one or two parameters, and they show how a simple mechanism can produce an important property seen in many real-world networks. For example, the preferential attachment model produces networks with a power-law degree distribution.

Other models are more parameter-intensive and they can produce realistic networks that exhibit more than one real-world properties. For example, the HRG is such a model and it can produce a hierarchical community structure, clustering, and a skewed degree distribution.

When using a network model – or potentially when you design your own network model – it is important to ask first: ***what is the goal of the model?***

A famous aphorism in science is that “all models are wrong but some are useful” (attributed to the statistician George Box).

So, asking whether “the model is realistic” is not usually the right question. The goal of a model is never to just copy reality (another aphorism in science is that “the best model of a cat is the same cat”). The goal of a model is to create one, or usually many, networks that exhibit certain properties.

We can use these networks to examine how the given properties affect the function of those networks. For example, we can examine how a family of networks with power-law degree distribution affect the spread of an epidemic.

There are 100s of network models in the literature – and of course, we could only study a small number of them. Nevertheless, this lesson gave you sufficient background to understand and use any other model you encounter in the future.